

SYLLABUS :-

Mechanics of deformable bodies (3-0-0, 42 lectures), No.NAXXXXXX Introduction to deformable bodies, stress-strain relation, compatibility. Bending of beams- load, shear and moment in beams, properties of cross sectional area, stresses in beams, equation of elastic curve, deflection. Energy Methods-minimum potential and complementary energy, Betti-Maxwell Reciprocal theorem. (18) Torsion of cylindrical bars, torsional stress, modulus of rigidity and deformation. Shear in beam (vertical, longitudinal shear, shear center in thin walled beam), Column-long and short, buckling of slender columns, Euler buckling load for different end conditions. (10) Transformation of stress and strain, principal stresses, principal strains, Mohr's circle for stress and strain, plain stress and plain strain problem, combined loading: axial and torsional; axial and bending; axial, torsional and bending. Pressure vessel, Hoop stress, Theories of failure- practical considerations for design, stress concentration, shear lag. (14) Texts: 1. Elements of strength of material by Timoshenko, Young 2. Engineering mechanics of solids by Popov 3. Mechanics of material by Beer, Johnston, DeWolf, Mazurek 4. Structural analysis by Hibbler 5. Advanced mechanics of solids by Srinath